

Mini segmentation: Dark spots of leaves infected by apple scab

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Introduction



- Data set
- Preparation for segmentation

Manual segmentation

Image: 1
(256x256x3, uint8)



Leaf mask
(256x256x3, uint8)

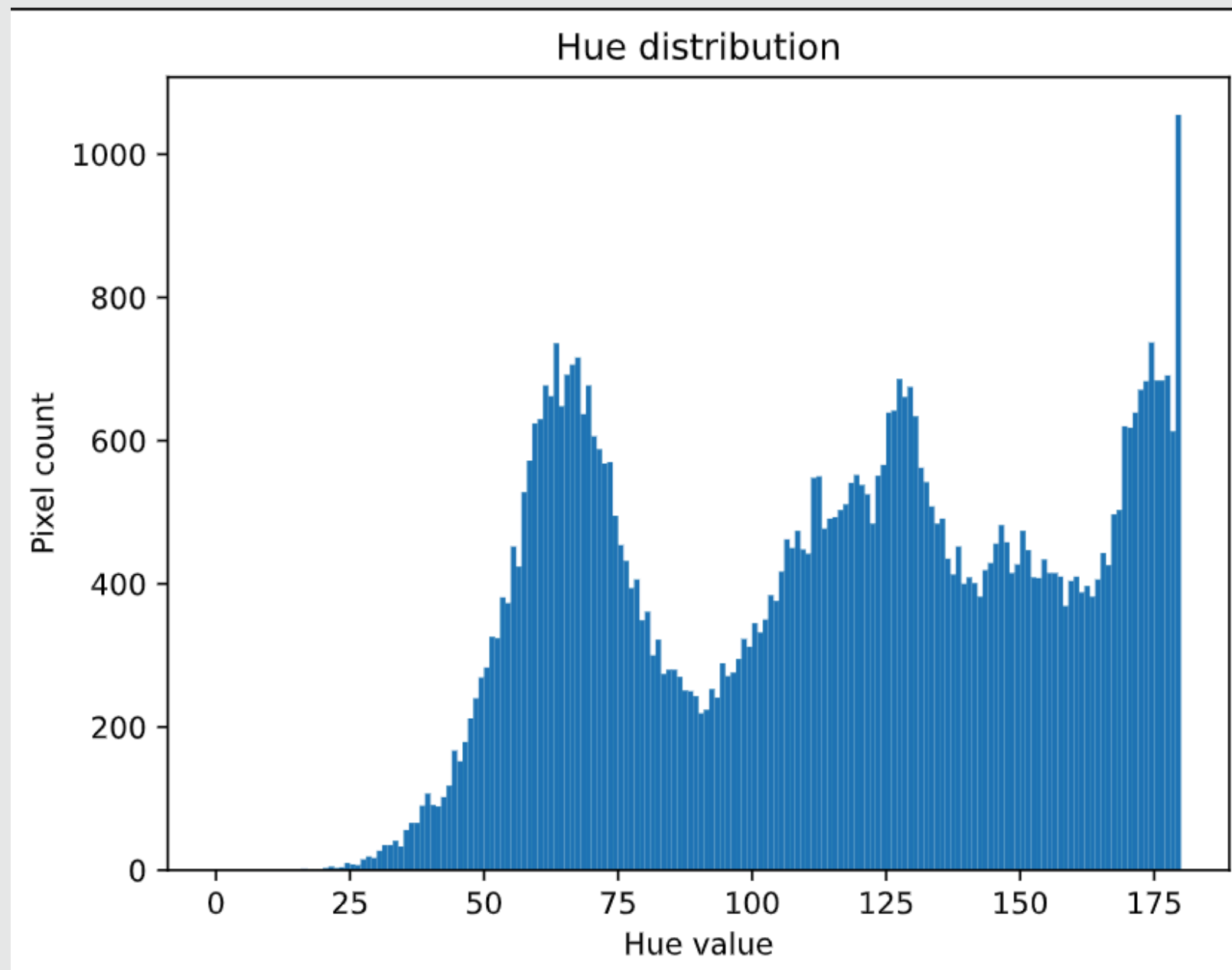


Spots mask
(256x256x3, uint8)



Automated segmentation

Channel samplig of dataset



Dark S range: 17 - 108

Dark V range: 43 - 125

Healthy S range: 51 - 134

Healthy V range: 86 - 149

Automated segmentation

Results

Image: 1
(256x256x3, uint8)



Leaf mask
(256x256, uint8)



Spots mask
(256x256, uint8)



Automated segmentation

Results

Image: 1
(256x256x3, uint8)



Leaf mask
(256x256, uint8)



Spots mask
(256x256, uint8)



Evaluation

=== Leaf masks: mean $\pm$ std ===

	Dice	IoU
mean	0.1463	0.0845
std	0.1339	0.0866

=== Dark spot masks: mean $\pm$ std ===

	Dice	IoU
mean	0.3651	0.2444
std	0.2178	0.1776

Discussion

- Differences in results between manual and automated segmentation.
- Channel distribution of RGB and HSV throughout the images.
- Morphological operations to enhance results.